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A Deep Dive into Portfolio Optimisation

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Abstract

Background: Selecting an optimal portfolio allocation across a universe of assets is a central problem in investment management. A wide range of allocation strategies have been proposed, from naive diversification to theoretically grounded optimisation frameworks, yet their relative out-of-sample performance remains contested, particularly across varying market regimes. This study empirically compares four portfolio strategies to assess whether model sophistication translates into superior investment performance.

Methodology: This study implements and compares four portfolio strategies: equal weight, Markowitz mean-variance, Black-Litterman and Risk Parity applied to an 18-asset UK multi-asset portfolio comprising 15 FTSE 100 equities, a UK government bond ETF, a gold ETC and a broad commodities ETF, over the period January 2015 to December 2025. Strategies are evaluated using a monthly rebalancing framework and assessed across annualised return, volatility, Sharpe ratio, maximum drawdown and portfolio turnover.

Results: We find results broadly consistent with DeMiguel et al. (2009), with the naive benchmark proving difficult to displace on a risk-adjusted basis over the full sample period. The optimisation strategies exhibit meaningfully different risk profiles, however, suggesting that Sharpe ratio alone does not fully capture the practical trade-offs between approaches.

1 Introduction

1.1 The Portfolio Allocation Problem

The portfolio allocation problem is the fundamental question in investment management. It asks: How should capital be distributed across available investment opportunities to best achieve an investor's objectives?

The problem was intractable until 1952, when Harry Markowitz published *Portfolio Selection* in the *Journal of Finance* and turned it into a clean optimisation problem, and the "efficient frontier" gave investors a principled answer. Markowitz's mean variance model minimises portfolio variance for a given level of expected return. It was revolutionary in the consideration of variance as the risk, standing as a hallmark of modern portfolio theory. In practice however, the model has proven difficult to implement reliably.

Michaud (1989) challenged this view directly, arguing that small estimation errors in expected returns produce wildly unstable portfolio recommendations.

In 2009, DeMiguel et al. delivered a provocative result: he found that even sophisticated strategies incorporating shrinkage estimators and Bayesian methods failed to consistently outperform 1/N out-of-sample, suggesting that estimation error dominates theoretical optimality for typical portfolio problems. This tension between the theoretical elegance of optimisation and its empirical fragility motivates the present study.

1.2 Research Objectives

This study addresses three main questions:

i. Do optimisation models outperform a naive 1/N benchmark out-of-sample, applied to a multi-asset UK portfolio over a ten-year period?
ii. Which model offers the best risk-adjusted performance, as measured by the Sharpe ratio?
iii. How does model performance vary across distinct market regimes: the Brexit referendum (2016), the COVID-19 crash (2020), and the 2022 rate-hiking cycle?

1.3 Structure of the Report

The remainder of this report is organised as follows:
Section 2 reviews the theoretical foundations of each model and derives their key mathematical results. Section 3 describes the data, implementation choices, and backtesting framework. Section 4 presents

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Key Points

- The equal-weight portfolio is robust because it does not rely on estimated parameters and is therefore unaffected by estimation error.
- The Efficient Frontier is the set of portfolios that deliver the minimum possible variance for each level of expected return.
- Empirical evidence suggests that many optimisation models fail to consistently outperform the equal-weight portfolio out of sample.

the empirical results, including full-period performance metrics and a breakdown by market regime. Section 5 discusses the findings and their practical implications. Section 6 concludes.

2 Literature Review and Background

2.1 Notation

Some common notation that is used across this paper is the following:

Symbol	Description
N	Number of assets in the portfolio
$\mathbf{w} \in \mathbb{R}^N$	Vector of portfolio weights
$\boldsymbol{\mu} \in \mathbb{R}^N$	Vector of expected returns
$\boldsymbol{\Sigma} \in \mathbb{R}^{N \times N}$	Covariance matrix of asset returns
$\mathbf{1} \in \mathbb{R}^N$	Vector of ones
$r_f \in \mathbb{R}$	Risk-free rate

2.2 The equal weight portfolio

The equal-weight (1/N) portfolio assigns an identical weight of 1/N to each asset in the portfolio, holding N assets, regardless of any asset-specific characteristics such as return, volatility, or correlation. DeMiguel et al. (2009) tested 14 optimisation models across 7 datasets. None consistently outperformed the naive equal-weight portfolio:

$$w_i = \frac{1}{N} \quad \text{for} \quad 1 \leq i \leq N \tag{2.1}$$

But why? When expected returns are estimated from historical data in order to determine optimal portfolio weights, the estimates improve as more data is collected, but they do so slowly.
Formally, the estimation error decays as

$$\hat{\mu} - \mu = O\left(\frac{1}{\sqrt{T}}\right) \tag{2.2}$$

where T is the number of observations.

Because the error shrinks at rate $\frac{1}{\sqrt{T}}$, halving the error requires four times as much data, meaning that with monthly data and realistic return distributions, approximately 500 years of observations are required for statistically reliable estimates, especially since these errors are amplified through the term $\Sigma^{-1}\hat{\mu}$.

The equal-weight portfolio sidesteps this problem entirely: by assigning a fixed weight to each asset, it requires no parameter estimation and is therefore immune to estimation error by construction.
The cost of this simplicity, however, is that the portfolio ignores all information about assets' risk and return characteristics, makes no attempt to diversify risk efficiently and cannot adapt to changing market conditions.

2.3 Markowitz's mean-variance optimisation

Markowitz (1952) formulated *Portfolio Selection* in 1952, approaching the asset allocation problem as a constrained optimisation problem: minimise portfolio variance for a given level of expected return, the foundation of the mean-variance model. His work stands as a hallmark of modern portfolio theory.

Formally, the problem states:

$$\min_{\mathbf{w}} \hat{\sigma} = \mathbf{w}^\top \hat{\boldsymbol{\Sigma}}\, \mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^\top \boldsymbol{\mu} = \mu_p \quad ; \quad \mathbf{w}^\top \mathbf{1} = 1 \tag{2.3}$$

with μ_p as the scalar target portfolio return.

The first constraint, also known as the return constraint, fixes the portfolio to a specific point at the target level μ_p, pinning the solution to a specific point on the "Efficient Frontier". By varying μ_p across all feasible values, the complete set of minimum-variance portfolios - the Efficient Frontier - is traced out. The second constraint, i.e. the budget constraint, ensures the weights represent a valid allocation of wealth, hence that the sum of all weights is always equal to 1.

It is important to note that no domain-restriction condition is imposed on the budget constraint, each w_i may take any real value, in particular, negative ones. Practically speaking, a negative weight corresponds to a short position in the associated asset.

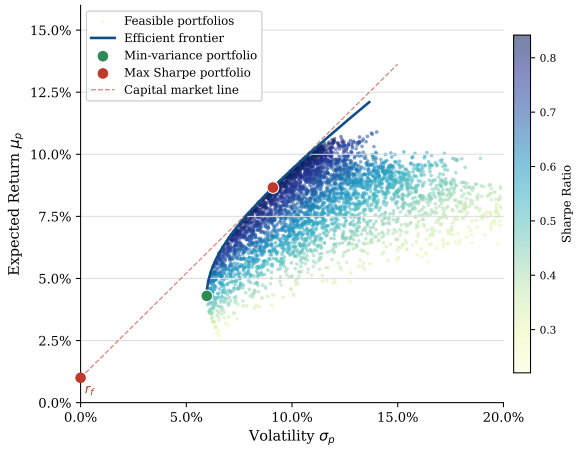

Figure 1. Efficient Frontier estimated using sample covariance

Because the objective function is quadratic and the constraints are linear equalities, we cannot solve the problem by direct substitution (two constraints, n unknowns). We therefore use the method of Lagrange multipliers. The Lagrangian formulation is:

$$\mathcal{L}(\mathbf{w}, \lambda, \gamma) = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} - \lambda\left(\mathbf{w}^\top \boldsymbol{\mu} - \mu_p\right) - \gamma\left(\mathbf{1}^\top \mathbf{w} - 1\right) \tag{2.4}$$

where λ and γ are the Lagrange multipliers associated with the return and budget constraints, respectively. Intuitively, λ captures the marginal cost of requiring a higher return (i.e. how much additional variance must be accepted per unit increase in μ_p), and γ enforces full investment.

Taking the first-order condition:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = 2\boldsymbol{\Sigma}\mathbf{w} - \lambda\boldsymbol{\mu} - \gamma\mathbf{1} = \mathbf{0} \tag{2.5}$$

Key Points

- Black-Litterman uses market-cap weights as a neutral starting point for expected returns.
- Subjective investor views are blended with the equilibrium prior through a Bayesian update.
- The optimisation step in Black-Litterman is the same as MVO with μ_{BL} replacing the sample mean.

Solving gives:

$$\mathbf{w}_p = \frac{\lambda}{2}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \frac{\gamma}{2}\boldsymbol{\Sigma}^{-1}\mathbf{1} \tag{2.6}$$

Imposing the budget and return constraints leads to the following two-fund theorem: all efficient portfolios can be expressed as a linear combination of any two distinct efficient portfolios.

$$\mathbf{w}_A = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1}} \qquad \mathbf{w}_B = \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}} \tag{2.7}$$

Note that $\mathbf{w}_A$ is the *global minimum variance* (GMV) portfolio, the leftmost point on the efficient frontier, obtained by dropping the return constraint and minimising variance subject only to full investment:

$$\mathbf{w}_{\text{GMV}} = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1}} \tag{2.8}$$

And the *maximum Sharpe ratio* portfolio is:

$$\mathbf{w}_{\text{MSR}} = \frac{\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f\mathbf{1})}{\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f\mathbf{1})} \tag{2.9}$$

Despite its theoretical elegance, Markowitz optimisation suffers from a well-documented problem. As mentioned above, the solution depends critically on $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, both estimated from historical data.

Small errors in μ can cause large shifts in optimal weights. Michaud (1989) characterised the optimiser as an "error maximiser": it systematically overweights assets with overestimated returns and underestimated variances.

2.4 Black-Litterman and its Bayesian approach

The Black-Litterman (BL) model was developed by Fischer Black and Robert Litterman at Goldman Sachs in the early 1990s to address the instability of Mean-Variance Optimisation (Black and Litterman; 1991, 1992).

The BL model addresses two shortcomings of classic mean-variance optimisation:

i. What expected returns should be used?
Rather than relying on historical data, BL uses returns implied by current market-cap weights as the neutral starting point.
ii. How should the views be incorporated?
BL offers a formal Bayesian framework for blending subjective investor views with this equilibrium prior, weighting each source inversely by its uncertainty.

Step 1. Equilibrium Returns

The model takes as its starting point the equilibrium expected returns implied by the market portfolio, i.e. the portfolio where each asset is held in proportion to its market capitalisation. Assuming the market portfolio is mean-variance efficient, the equilibrium excess returns are:

$$\Pi = \delta\Sigma w_m \tag{2.10}$$

where w_m is the market-cap weight vector, Σ is the covariance matrix of asset returns and δ is the risk-aversion coefficient

$$\delta = \frac{r_m - r_f}{\sigma_m^2} \tag{2.11}$$

with r_m the return of the market portfolio and σ^2 its variance.

Step 2. Express Views

A view can be absolute or relative, for example "asset A will return 5%" or "asset A will outperform asset B by 2%". Each view has two components: the expected return itself and a confidence level attached to it.

Formally, k views are expressed through the following system:

$$P\boldsymbol{\mu} = \boldsymbol{Q} + \boldsymbol{\varepsilon} \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \Omega) \tag{2.12}$$

where:

Symbol	Description
$P \in \mathbb{R}^{k\times n}$	*Pick matrix*: encodes which assets views involve
$\boldsymbol{\mu} \in \mathbb{R}^k$	Vector of expected returns
$\boldsymbol{Q} \in \mathbb{R}^k$	Expected view returns
$\Omega \in \mathbb{R}^{k\times k}$	View uncertainty matrix (diagonal)
$\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \Omega)$	View error: deviation of true returns from stated views

Assets are ranked by their return over the previous 12 months, excluding the most recent month to eliminate short-term reversal effects. The highest-return tercile forms the winner portfolio, while the lowest-return decile constitutes the loser portfolio.

The view is expressed as:

$$\boldsymbol{p}_k^\top\boldsymbol{\mu} = q_k \tag{2.13}$$

$\boldsymbol{p}_k$ assigns equal positive weights to winner assets and equal negative weights to loser assets, forming a long-short momentum view. The expected return spread q_k is estimated over a rolling window at each rebalancing date as asset rankings change.

Step 3. Bayesian Update

The prior on expected returns is:

$$\boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\Pi}, \tau\boldsymbol{\Sigma}) \tag{2.14}$$

This encodes the belief that expected returns are centred on the equilibrium vector $\boldsymbol{\Pi}$, with uncertainty scaled by τ. Combining this prior with the view system via Bayes' theorem yields the posterior distribution over expected returns:

$$\boldsymbol{\mu} \mid \boldsymbol{Q} \sim \mathcal{N}(\boldsymbol{\mu}_{BL}, \boldsymbol{\Sigma}_{BL}) \tag{2.15}$$

where the posterior mean is:

$$\boldsymbol{\mu}_{BL} = \left[(\tau\Sigma)^{-1} + P^\top\Omega^{-1}P\right]^{-1}\left[(\tau\Sigma)^{-1}\boldsymbol{\Pi} + P^\top\Omega^{-1}\boldsymbol{Q}\right] \tag{2.16}$$

The posterior mean $\boldsymbol{\mu}_{BL}$ is a precision-weighted average of the equilibrium prior $\boldsymbol{\Pi}$ and the investor's views $\boldsymbol{Q}$, with each source weighted inversely by its uncertainty. When views are diffuse (large $\boldsymbol{\Omega}$), $\boldsymbol{\mu}_{BL}$ remains close to $\boldsymbol{\Pi}$. If the views are held with high confidence

Key Points

- Risk Parity allocates capital such that every asset contributes equally to total portfolio volatility, not equally to capital.
- Unlike MVO and Black-Litterman, Risk Parity requires no estimate of expected returns: allocations depend only on the covariance structure.
- The equal risk contribution condition $w_i(\boldsymbol{\Sigma}\mathbf{w})_i = c$ for all i has no closed-form solution and in general and is solved numerically.

(small $\boldsymbol{\Omega}$), $\boldsymbol{\mu}_{BL}$ tilts toward $\mathbf{Q}$.

This blending property is central to the appeal of the BL framework: Rather than overriding the equilibrium prior, views are incorporated gradually and in proportion to their confidence. This acts as a natural regulariser, because the prior pulls weights toward the market portfolio, the model is far less prone to the concentrated, unstable allocations that plague classical MVO.

Step 4. Portfolio Optimisation

The optimal portfolio is obtained by applying the same Sharpe-maximising objective derived in Section 2.3 (equation 2.9), substituting $\boldsymbol{\mu}_{\mathrm{BL}}$ in place of the sample mean $\boldsymbol{\mu}$.

2.5 Risk Parity

The modern Risk Parity framework was popularised by Ray Dalio at Bridgewater Associates and underpins the All Weather strategy introduced in the 1990s. The approach was subsequently formalised by Maillard et al. (2010), who derived the mathematical conditions for equal risk contribution portfolios.

The central motivation departs entirely from the Mean Variance and Black-Litterman frameworks: rather than optimising expected return for a given level of risk, Risk Parity ignores return estimates altogether and instead focuses on how risk is distributed across the portfolio.

Risk Parity addresses a structural limitation shared by both MVO and the equal-weight portfolio:

i. In MVO and BL, allocations depend on estimated expected returns, which are noisy and prone to estimation error.
ii. In equal-weight portfolios, capital is diversified equally but risk is not. Assets with high volatility dominate total portfolio risk despite receiving the same capital allocation.

Risk Decomposition

Portfolio volatility $\sigma_p = \sqrt{\mathbf{w}^\top \boldsymbol{\Sigma}\mathbf{w}}$ is a homogeneous function of degree 1 in $\mathbf{w}$. By Euler's homogeneous function theorem, it admits the following decomposition:

$$\sigma_p = \sum_{i=1}^{N} w_i \frac{\partial \sigma_p}{\partial w_i} \tag{2.17}$$

The *marginal risk contribution* (MRC) of asset i is:

$$\mathrm{MRC}_i = \frac{\partial \sigma_p}{\partial w_i} = \frac{(\boldsymbol{\Sigma}\mathbf{w})_i}{\sigma_p} \tag{2.18}$$

and the total *risk contribution* (RC) of asset i is:

$$\mathrm{RC}_i = w_i \cdot \mathrm{MRC}_i = \frac{w_i\,(\boldsymbol{\Sigma}\mathbf{w})_i}{\sigma_p} \tag{2.19}$$

with the useful property that $\sum_{i=1}^{N} \mathrm{RC}_i = \sigma_p$.

The Risk Parity Condition

The portfolio is said to be *risk-balanced* when every asset contributes equally to total portfolio volatility:

$$\mathrm{RC}_i = \frac{\sigma_p}{N} \quad \forall\, i \tag{2.20}$$

Since σ_p appears on both sides, this reduces to the equivalent condition:

$$w_i\,(\boldsymbol{\Sigma}\mathbf{w})_i = c \quad \forall\, i \tag{2.21}$$

for some constant $c > 0$, subject to $\mathbf{1}^\top \mathbf{w} = 1$ and $\mathbf{w} \geq \mathbf{0}$. This system has no closed-form solution in general and is solved numerically; the implementation details are described in Section 3.

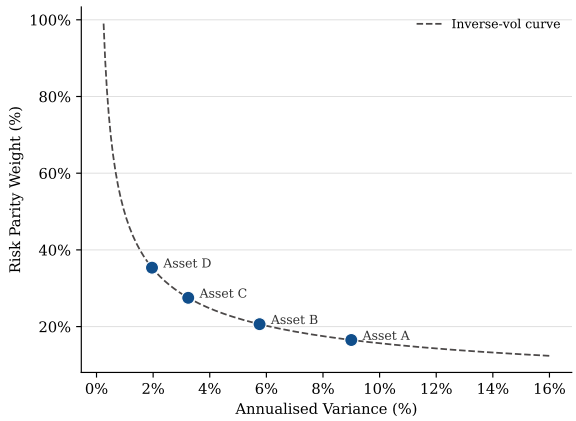

Figure 2. Risk parity weights as a function of asset variance

Special Case: Uncorrelated Assets

When $\boldsymbol{\Sigma}$ is diagonal, the risk parity condition reduces to an analytic solution. Equal risk contribution then requires each asset's weight to be proportional to the inverse of its volatility:

$$w_i^{\mathrm{RP}} = \frac{1/\sigma_i}{\sum_{j=1}^{N} 1/\sigma_j} \tag{2.22}$$

This inverse-volatility weighting is the closed-form limit of risk parity, and also serves as the natural initialisation point for the numerical solver in the general case.

Despite its appeal, Risk Parity has a well-known structural consequence: it mechanically overweights low-volatility assets. In a portfolio containing both equities and government bonds, bonds receive substantially larger capital allocations precisely because their volatility is lower, which raises the question of whether the resulting portfolio is truly diversified in an economic sense or merely in a statistical one.

3 Methodology

3.1 Data

The dataset comprises daily adjusted closing prices for 18 assets over an 11-year period from 1 January 2015 to 31 December 2025, sourced programmatically from *Yahoo Finance*. The universe is constructed to reflect a representative UK-centric multi-asset portfolio: 15 large-cap FTSE 100 equities spanning eight sectors, supplemented by three exchange-traded products providing exposure to UK government bonds, gold, and

Key Points

- The universe comprises 18 assets across diversified sectors, including equities, bonds, gold and commodities.
- Monthly rebalancing on the final trading day of each calendar month is applied consistently across all four strategies.
- The walk-forward design of the backtester enforces a strict information barrier, ensuring no look-ahead bias.

broad commodities. The full asset universe is listed in Table 1.

Table 1. Asset universe

Asset Class	Name	Ticker
Banking	HSBC	HSBA.L
	Lloyds Banking Group	LLOY.L
	Barclays	BARC.L
Oil & Gas	Shell	SHEL.L
	BP	BP.L
Consumer Goods	Unilever	ULVR.L
	Tesco	TSCO.L
	Diageo	DGE.L
Pharmaceuticals	AstraZeneca	AZN.L
	GSK	GSK.L
Mining	Rio Tinto	RIO.L
	Glencore	GLEN.L
Utilities	National Grid	NG.L
Telecoms	Vodafone	VOD.L
Industrials	Rolls-Royce	RR.L
UK Govt Bonds	iShares Core UK Gilts ETF	IGLT.L
Precious Metals	Invesco Physical Gold ETC	SGLD.L
Commodities	WisdomTree Broad Commodities	WCOG.L

Adjusted closing prices are used in preference to raw prices. Yahoo Finance's adjusted close corrects for corporate actions including stock splits, rights issues, and dividend distributions, ensuring that computed returns reflect the true total return experienced by an investor. Missing observations arising from non-synchronous trading calendars are handled by forward-filling the most recent available price prior to return calculation.

The risk-free rate is proxied by the 3-month UK Treasury Bill rate, sourced from the Federal Reserve Economic Data database and converted to a daily rate *FRED Database*. The rate is applied time-variably across the sample, capturing the material shift from near-zero rates during 2015–2021 to above 5% during the 2023-2024 tightening cycle.

3.2 Model Implementation

All models are implemented from first principles in Python using `NumPy` and `Pandas`, following an object-oriented design in which each strategy inherits from a common abstract base class. This ensures full transparency and reproducibility of all optimisation steps. The complete source code is publicly available at the project repository (Surrey Capital Research; 2026).

All models share a rolling estimation framework: at each monthly rebalancing date, covariance and return inputs are re-estimated using the preceding 252 trading days of data. Table 2 summarises the key implementation parameters for each strategy.

Equal Weight assigns $w_i = 1/N$ at each rebalancing date and requires no parameter estimation.

In the Mean-Variance model, the maximum Sharpe ratio portfolio is solved via an active-set method: assets with negative target weights are iteratively pruned from the investment set and the system is re-solved until all remaining weights satisfy the long-only constraint $w_i \geq 0$. A small ridge term $\lambda = 10^{-8}$ is added to the diagonal of $\hat{\boldsymbol{\Sigma}}$ at each step to guarantee numerical positive definiteness.

Table 2. Implementation parameters by strategy

Model	Window	Constraints	Solver
Equal Weight	None	$w_i = 1/N$	Closed-form
MVO	252 days	Long-only	Active-set
Black-Litterman	252 days	Long-only	Active-set
Risk Parity	252 days	Long-only	Newton-Raphson

Risk Parity solves the equal risk contribution system (equation 2.21) via Newton-Raphson iteration, initialised with inverse-volatility weights. Convergence is typically achieved within 10 iterations.

The Black-Litterman model requires two calibration parameters. The **prior uncertainty scalar** $\tau \in (0, 1)$ controls how much uncertainty surrounds the equilibrium prior: a small τ implies strong confidence that expected returns are close to $\boldsymbol{\Pi}$, assigning little weight to the views. We set $\tau = 0.05$, consistent with the original implementation of He and Litterman (2002), who argued that uncertainty about the prior should be considerably smaller than uncertainty about the returns themselves.

The **risk-aversion coefficient** δ determines the magnitude of the equilibrium returns via $\boldsymbol{\Pi} = \delta \boldsymbol{\Sigma} \mathbf{w}_m$; a higher δ implies investors require greater compensation per unit of portfolio variance, producing larger implied equilibrium returns. We set $\delta = 3$, a standard value in the BL literature (Walters; 2011). A sensitivity analysis varying both parameters across a grid is reported in Section 5.3.

The investor views incorporated are motivated by the momentum effect documented by Jegadeesh and Titman (1993), showing that stocks with high returns over the prior 3-12 months continue to outperform stocks with low prior returns over the subsequent 3-12 months, generating abnormal returns of approximately 1% per month. This persistent pattern offers an empirically grounded basis for constructing views within the BL framework.

A long-short view is then formed between the top and bottom terciles. The posterior expected returns $\boldsymbol{\mu}_{BL}$ are passed into the same active-set MVO solver used for the standard MVO strategy.

3.3 Backtesting Framework

The four strategies are evaluated using a walk-forward backtesting engine to ensure zero look-ahead bias. The engine enforces a strict information barrier by passing only prices up to the rebalancing date; the covariance dependent strategies (MVO, BL, RP) then estimate over a 252 trading day sub-window. This distinction is already implicit in Table 2 (Section 3.2).

A starting capital of £100,000 is deployed on the first available trading day. The portfolio is thereafter rebalanced on the final trading day of each calendar month, with all positions revalued daily at the adjusted closing price.

The simulation assumes perfect asset divisibility (i.e. the existence of fractional shares) and immediate execution at the closing price on each rebalancing date. Transaction costs are not modelled; portfolio

Key Points

- Sharpe and Sortino use annualised arithmetic mean return μ_p, not CAGR, in the numerator.
- Maximum drawdown measures the largest peak-to-trough loss in portfolio value over the sample.
- Average monthly turnover quantifies rebalancing intensity and serves as a proxy for implementation cost.

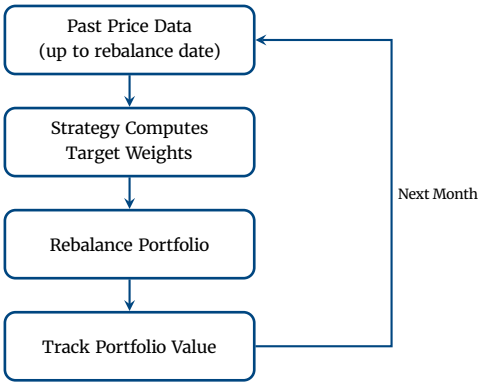

Figure 3. Walk-forward backtesting loop

turnover is instead reported as a standalone metric, allowing the reader to assess the practical trading burden of each strategy independently.

Bid-ask spreads and market impact are not accounted for; given that all 18 assets are large-cap and exchange-traded, it is reasonable to assume sufficient liquidity to absorb the portfolio's order flow without material execution slippage.

Each simulation run produces a `BacktestResult` object containing the daily equity curve, full position history and a trade log. Performance metrics are computed from the equity curve at the conclusion of each run and are defined in Section 3.4.

3.4 Performance Metrics

Nine metrics are used to evaluate each strategy, grouped into three categories: return, risk, and risk-adjusted. Let $r_t = V_t/V_{t-1} - 1$ denote the daily portfolio return, V_0 and V_T the initial and terminal portfolio values, T the total number of trading days, $\mu_p = \bar{r}_t \times 252$ the annualised arithmetic mean return, and r_f the mean annualised risk-free rate over the sample. Table 3 defines each metric formally.

Return metrics capture the magnitude of wealth creation. CAGR annualises the compounded total return to allow comparison across strategies regardless of sub-period variation.

Risk metrics characterise the severity and shape of losses: maximum drawdown measures the worst peak-to-trough decline an investor would have experienced; VaR and CVaR characterise tail losses from the empirical return distribution without parametric assumptions.

Risk-adjusted metrics normalise excess return by some measure of risk: the Sharpe ratio uses total volatility, while the Sortino ratio penalises only downside deviations, making it more appropriate when return distributions are asymmetric.

VaR and CVaR are computed from the empirical distribution of daily returns without a parametric assumption. $\hat{\sigma}_{\text{down}}$ denotes the annualised standard deviation of negative daily returns only, so the Sortino ratio penalises downside volatility exclusively.

Average monthly turnover measures the mean total absolute weight change across all assets at each rebalancing date, providing a strategy-level indicator of trading activity.

Table 3. Performance metrics used to evaluate each strategy

Metric	Formula	Category
Total Return	$V_T/V_0 - 1$	Return
CAGR	$(V_T/V_0)^{252/T} - 1$	Return
Volatility	$\hat{\sigma}(r_t) \cdot \sqrt{252}$	Risk
Max Drawdown	$\min_t \left(\frac{V_t}{\max_{s \le t} V_s} - 1 \right)$	Risk
95% VaR	$\text{Percentile}(r_t,\ 5\%)$	Risk
95% CVaR	$\mathbb{E}[\, r_t \mid r_t \le \text{VaR}\,]$	Risk
Sharpe Ratio	$(\mu_p - r_f)\ /\ \hat{\sigma}_p$	Risk-Adjusted
Sortino Ratio	$(\mu_p - r_f)\ /\ \hat{\sigma}_{\text{down}}$	Risk-Adjusted
Avg. Monthly Turnover	$\frac{1}{M} \sum_{m=1}^{M} \sum_{i=1}^{N} \lvert w_{i,m} - w_{i,m-1} \rvert$	Portfolio

4 Results

4.1 Full-Period Performance

Tables 4 and 5 present the return, risk, and downside risk metrics for all four strategies over the full sample period from 01/2015 to 12/2025, with Figure 3 showing the corresponding cumulative wealth curves and Figure 4 the drawdown profiles.

Table 4. Return and Risk Metrics (2015-2025)

Metric	Equal Weight	Risk Parity	MVO	BL
Total Return	93.9%	41.2%	35.7%	62.7%
CAGR	6.2%	3.2%	2.8%	4.5%
Volatility	14.6%	10%	11.1%	13.5%
Sharpe Ratio	0.36	0.2	0.15	0.26
Max Drawdown	-30.6%	-23.7%	-23.7%	-27.2%

Table 5. Downside Risk Metrics (2015-2025)

Metric	Equal Weight	Risk Parity	MVO	BL
Sortino Ratio	0.47	0.23	0.18	0.35
95% VaR	-1.4%	-0.9%	-1.1%	-1.2%
95% CVaR	-2.2%	-1.6%	-1.7%	-2.1%

Raw returns

Equal Weight delivered the strongest total return over the full period at 93.9% (CAGR: 6.2%), with its portfolio value reaching approximately £194k by end-2025. Black-Litterman was the second-best strategy at 62.7% (CAGR: 4.5%), followed by Risk Parity at 41.2% and MVO at 35.7%, the weakest in absolute terms. The three optimised strategies finished between £136k and £163k, well below the naive $1/N$ benchmark.

Risk-adjusted performance

The ranking is preserved on a risk-adjusted basis. Equal Weight retains the top position with a Sharpe ratio of 0.36 and Sortino of 0.47, reflecting that its higher volatility (14.55%) was more than compensated by superior returns. Black-Litterman comes second (Sharpe: 0.26, Sortino: 0.35), followed by Risk Parity (0.19 / 0.23) and MVO (0.15 / 0.18).

Key Points

- Equal Weight achieves the highest total return (93.9%) and Sharpe ratio (0.36) of all four strategies.
- No strategy dominates uniformly: performance rankings shift materially across market regimes.
- MVO generates the highest monthly turnover (34.7%), over 13 times that of Equal Weight (2.5%).

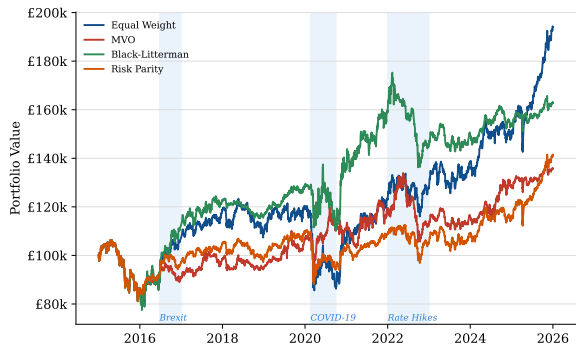

Figure 4. Cumulative wealth curves for all four strategies, indexed to £100,000 initial capital

Notably, all three optimised strategies underperformed the naive Equal Weight benchmark on a risk-adjusted basis: A well-documented result known as the **"1/N puzzle"**. In practice, estimation error in expected returns and covariances often causes optimised portfolios to underperform simpler allocations out-of-sample.

The cumulative returns chart presents a broadly upward narrative, but the drawdown chart tells a more nuanced story about the cost of participating in that growth.

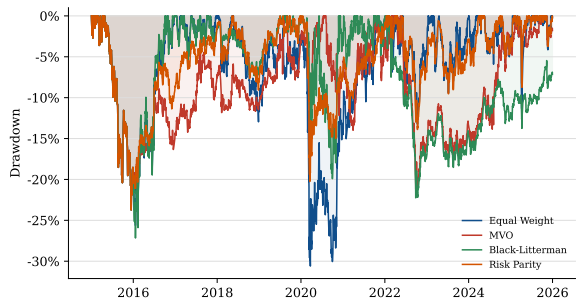

Figure 5. Drawdown profiles

Equal Weight's hidden fragility

Despite delivering the highest returns, Equal Weight also experienced the largest maximum drawdown at −30.6%, most visibly during the COVID-19 market crash in early 2020. At that point, a £100,000 portfolio would have temporarily lost over £30,000 of value, a risk that is easily missed in the upward trajectory of the cumulative chart.

Risk Parity's resilience under stress

Risk Parity and MVO share the smallest max drawdown (−23.7%), and the drawdown chart in Figure 5 shows Risk Parity recovered relatively quickly from most dips. Its comparatively low volatility (10%) results in drawdowns that are both milder and shorter in duration, which is valuable for investors with lower loss tolerance or shorter time horizons.

Black-Litterman's 2022-2024 struggle

The drawdown chart highlights a sustained decline reaching −27.2% between 2022 and 2024. This risk episode is partially masked in the cumulative return plot due to the strong performance recorded in 2020-2022.

4.2 Risk-Return Trade-off

Figures 6 and 7 examine where each strategy sits in risk-return space and how risk-adjusted performance evolved over time.

Risk-return positioning

Equal Weight sits in the upper-right of the scatter, combining the highest return with the highest volatility (14.6%). Despite this, it clears the 0.3 Sharpe isocurve, confirming it as the best risk-adjusted strategy of the four.

Risk Parity occupies the opposite corner: the lowest volatility (10.0%) but also the lowest return, placing it just above the 0.1 Sharpe isocurve.

Black-Litterman sits at nearly the same volatility as Equal Weight (13.5%) but with a return more than 1.5 percentage points lower, offering poor compensation for the risk taken.

MVO is the most difficult to justify: it achieves lower return than Risk Parity yet higher volatility, placing it in a dominated position in risk-return space.

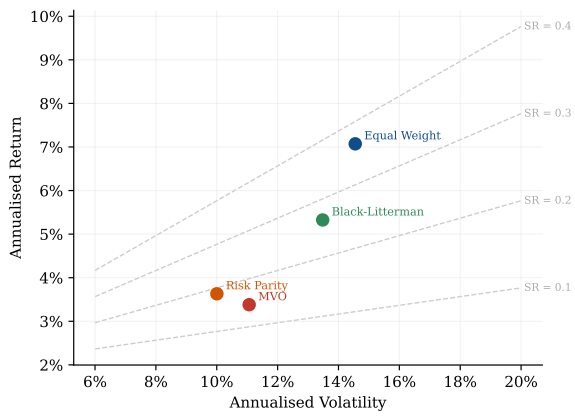

Figure 6. Risk-return scatter, dashed lines are Sharpe ratio isocurves

Performance consistency over time

The rolling 12-month Sharpe ratio reveals that no strategy maintains a persistent advantage across the full sample. All four strategies move largely in tandem - rising together through the 2016-2019 period, collapsing sharply in the COVID crash, recovering into 2021 and then compressing again through the 2022 rate-hiking cycle.

This co-movement suggests that the macroeconomic regime is a stronger determinant of short-term risk-adjusted performance than the choice of allocation model. The implication is direct: model selection alone does not insulate a portfolio against systemic shocks, which motivates the episode-by-episode analysis in Section 4.3.

Key Points

- BL excelled during the Brexit shock (+20.3%); MVO was uniquely resilient during COVID (+3.6%).
- All strategies declined in 2022 as simultaneous equity and bond falls broke diversification logic.
- Rolling Sharpe co-movement confirms macroeconomic regime dominates short-term model differences.

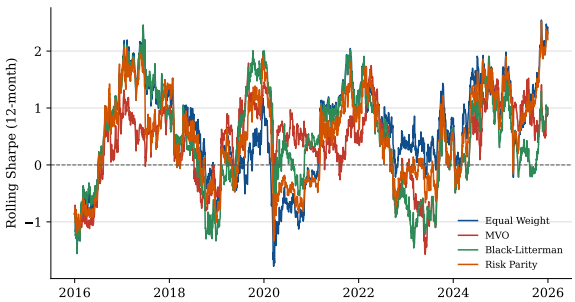

Figure 7. Rolling 12-month Sharpe ratio

Figure 8. Regime: Brexit (Jun-Dec 2016)

4.3 Regime Dependence

Full-period metrics can mask very different behaviour across market conditions. This section stress-tests the four strategies across three distinct episodes: the Brexit shock (Jun-Dec 2016), the COVID-19 crash (Feb-Sep 2020) and the 2022 rate-hiking cycle (Jan-Dec 2022), each representing a different type of market disruption. Table 6 and Figure 4 summarise cumulative returns across these periods.

Table 6. Strategy Return During Key Market Events

Event	Equal Weight	Risk Parity	MVO	BL
Brexit	15.9%	6.2%	0.8%	20.3%
COVID	-25.8%	-13.7%	3.6%	-13.4%
Rate Hikes	1.5%	-4.3%	-6%	-10.9%

Brexit (Jun-Dec 2016)

The Brexit referendum triggered a sharp sterling depreciation and a rapid rotation into exporters and internationally-exposed equities. Black-Litterman was the standout performer, gaining 20.3% over the period. Its view-adjusted allocations appear to have captured the post-referendum equity rebound effectively.

Equal Weight followed at 15.9%, benefiting from its broad exposure across assets without concentration risk pulling it in the wrong direction.

Risk Parity lagged considerably at 6.2%, its defensive, risk-equalising stance limited upside participation during a strong directional move.

MVO was essentially flat at 0.8%, suggesting its concentrated positioning was poorly aligned with the assets that benefited most from the shock.

COVID-19 (Feb-Sep 2020)

The COVID crash was sudden and indiscriminate, hitting all strategies hard in March 2020. What made a real difference is how quick and complete their recovery was.

MVO recovered fastest and was the only strategy to finish the period in positive territory (+3.6%), its concentrated positioning in two safe-haven assets: gold and UK Gilts, proved well-suited to a crash driven by a flight to safety.

Black-Litterman and Risk Parity ended the period in almost identical territory (−13.4% and −13.7% respectively), but went under different paths. Black-Litterman clawed its way back to breakeven by June 2020 before a second selloff dragged it back down, while Risk Parity simply never recovered, staying underwater for the entire period without any meaningful rebound.

Equal Weight was the worst performer by far, ending down −25.8%. Its broad equity exposure amplified the drawdown, and the chart shows it had not recovered anywhere near its starting level by October 2020.

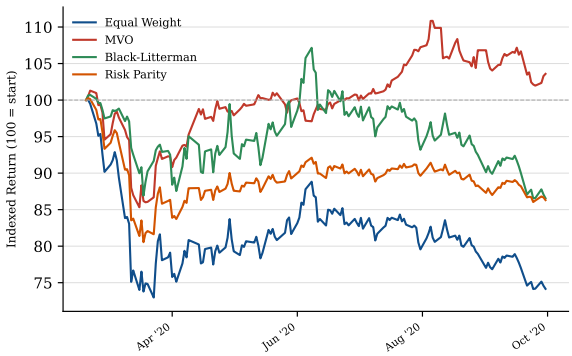

Figure 9. Regime: COVID-19 (Feb-Sep 2020)

Rate Hikes (Jan-Dec 2022)

2022 was uniquely damaging: equities and bonds sold off together, breaking a well-established negative correlation between these assets.

Equal Weight was the most resilient, ending the year roughly flat at +1.5%. Its lack of explicit bond exposure weighting meant it was less exposed to the bond sell-off than the more sophisticated models.

Risk Parity (−4.3%) held up better than its structural weakness would suggest. While the equity-bond correlation breakdown directly undermines its core diversification logic, losses were contained to a level that outperformed both MVO and Black-Litterman. This suggests that the risk spreading across other assets provided a cushion for the strategy.

MVO (−6%) fared worse than Risk Parity despite having no structural sensitivity to bond markets. Black-Litterman was the worst performer at −10.9%, its view-adjusted allocations, effective during Brexit, proved ill-suited to the speed and severity of the 2022 repricing.

Key Points

- MVO's average monthly turnover of 34.7% is over 13 times that of Equal Weight (2.5%).
- Risk Parity maintains the most stable allocations (ignoring equal weight), consistent with its covariance-only objective.
- Transaction costs not modelled here would disproportionately erode MVO's already weakest gross returns.

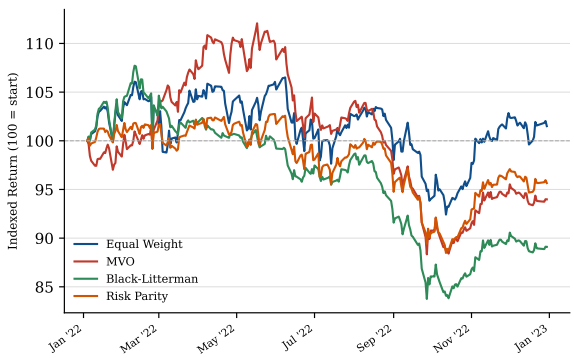

Figure 10. Regime: Rate Hikes (Jan-Dec 2022)

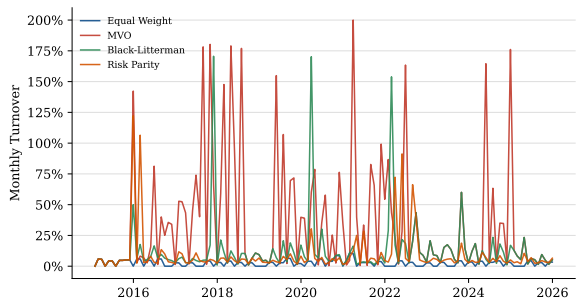

Figure 11. Monthly portfolio turnover for all four strategies. Each data point is the total absolute weight change across all assets at that month's rebalancing date

4.4 Portfolio Characteristics

Allocation stability

The three optimised strategies differ substantially in how they distribute and maintain their allocations over time. MVO is the most unstable: concentration rotates between a small number of assets, most notably IGLT.L and SGLD.L while the majority of the portfolio sits near zero for extended stretches. This "all or nothing" behaviour is characteristic of mean-variance optimisation, which tends to produce corner solutions that change dramatically from one rebalancing date to the next.

Black-Litterman is smoother, with weights more evenly distributed and less violent rotation, though IGLT.L consistently attracts a large allocation from 2018 onwards: a concentration risk that partially undermines the diversification the model is designed to provide.

Risk Parity is the most stable: allocations are broadly uniform across assets for most of the period, consistent with its objective of equalising risk contribution rather than capital. The persistent overweight in IGLT.L reflects its low historical volatility (6%), which mechanically attracts a larger risk budget within the framework.

Turnover and the cost of complexity

High turnover is not a sign of a well-functioning strategy, it is a cost. Figure 11 shows MVO trading at an average monthly rate of 34.7%, by far the highest of the four: more than twice Black-Litterman (12.7%), nearly four times Risk Parity (9%), and almost fourteen times Equal Weight (2.5%). The chart also reveals that MVO's turnover is not constant but spikes sharply at certain rebalancing dates, particularly around regime shifts, reflecting its sensitivity to changes in the estimated return and covariance inputs.

Every rebalancing generates transaction costs like brokerage fees and bid-ask spreads that erode net returns. Given that MVO already delivers the weakest gross performance, its net returns after implementation costs would be weaker still. Equal Weight, by contrast, requires minimal intervention yet outperforms all three optimised strategies over the full period. This is perhaps the most pointed finding of the section: **complexity does not necessarily pay.**

5 Discussion

5.1 Why did 1/N perform well?

The headline finding of this study is that Equal Weight, a strategy requiring no parameter estimation and no optimisation, delivered the highest cumulative return, the highest Sharpe ratio and the highest Sortino ratio of all four strategies over the full 2015-2025 period. This is not a marginal result: it outperformed the next-best strategy (Black-Litterman) by over 31 percentage points in total return and by 0.1 Sharpe units.

The natural question is: Why do three strategies that explicitly attempt to exploit information about returns and covariances consistently fail to improve on a naive benchmark?

Estimation error as the root cause

The core explanation is estimation error. Mean-variance optimisation and its derivatives require two sets of inputs: a vector of expected returns μ and a covariance matrix Σ. With 18 assets, this amounts to 18 mean estimates and 171 distinct covariance entries (189 parameters in total) each estimated from a rolling window of approximately 252 daily observations.

The statistical precision of these estimates is inherently limited: sample means estimated from one year of data carry enormous standard errors, and small perturbations in μ translate into large changes in the optimiser's output. Michaud (1989) characterised this as the "error maximisation" problem: the optimiser treats estimation noise as genuine signal and concentrates heavily into the assets whose returns happened to look most attractive over the estimation window, regardless of whether that outperformance will persist.

Equal Weight sidesteps this entirely by assuming no asset is expected to outperform any other: a prior that, on short rolling windows, turns out to be approximately correct.

An established empirical regularity

This finding is not unique to our dataset. DeMiguel et al. (2009) conducted a systematic comparison of 14 portfolio optimisation strategies across seven empirical datasets and found that none significantly outperformed 1/N on a risk-adjusted basis out-of-sample.

They derived a theoretical condition under which MVO improves on EW: the estimation window must exceed $N(N+1)/2$ periods, which for 18 assets implies over 170 years of data; far beyond any realistic investment horizon. Our results replicate their finding on

Key Points

- Risk Parity's structural overweight in low-volatility gilts drags returns during equity bull markets.
- The 2022 rate-hiking cycle broke the equity-bond diversification logic that underpins the strategy.
- Unlevered risk parity lacks the return compensation that leveraged institutional implementations achieve.

a UK multi-asset universe with a ten-year sample period. The 1/N puzzle is therefore not a curiosity of our particular dataset, but a well-documented property of mean-variance optimisation in realistic finite-sample settings.

Why this universe is particularly amenable to EW

The composition of the 18-asset portfolio matters. Because the universe spans genuinely distinct risk classes: UK equities, government bonds, gold, and broad commodities, a naive equal-weight allocation achieves meaningful cross-asset diversification without requiring any return forecast.

Had the universe consisted of 18 highly correlated stocks within a single sector, the free lunch of naive diversification would be substantially smaller, and the marginal value of optimisation would be greater. The multi-asset structure also means that EW's equity overweight relative to bonds happened to pay off during the prolonged equity bull market of 2016-2021, a period which accounts for the majority of the performance gap.

Implications for portfolio construction

These results should not be read as a rejection of portfolio optimisation theory. The Markowitz framework and its Bayesian extensions are theoretically coherent; the issue is that their practical implementation with short estimation windows and no regularisation produces unstable, concentrated portfolios that perform poorly out-of-sample.

Black-Litterman was designed precisely to address this by anchoring expected returns to a stable equilibrium prior and updating it only partially in the direction of investor views, thereby reducing sensitivity to noisy historical return estimates. BL's underperformance in this study reflects the specific calibration choices made, in particular the dominance of momentum views over the equilibrium prior, rather than a fundamental deficiency in the Bayesian approach. This point is developed further in Section 5.3.

5.2 The risk parity puzzle

Risk Parity recorded the lowest volatility (10%), the joint-smallest maximum drawdown (−23.7%), and the most stable allocation profile of any strategy in this study. By the conventional logic of risk-adjusted performance, a portfolio that controls downside more effectively should produce a competitive Sharpe ratio.

Instead, Risk Parity delivered the lowest Sharpe (0.19) and the lowest total return (41.2%) of the three optimised strategies, a result that warrants explanation.

Structural bond overweight

The root cause lies in how risk parity assigns capital. Because the strategy equalises risk contributions rather than weights, assets with low historical volatility mechanically receive a disproportionately large allocation. IGLT.L, the UK gilts ETF, carries approximately one-third the volatility of the equity constituents (6% versus 18-25%), and consequently attracts a far larger risk budget than any individual equity.

Over the 2015-2025 period, UK government bonds delivered modest nominal returns, and the strategy was structurally underweight in the equities that drove the majority of cross-asset returns.

This is not a calibration error or an implementation flaw but it is the defining consequence of risk parity on a multi-asset universe in which equities dominate risk-adjusted performance. The strategy achieves what it promises: equal risk contribution. The cost is a persistent drag on returns during equity bull markets.

Correlation breakdown in the 2022 regime

Risk parity's theoretical appeal rests on the historically low or negative correlation between equities and government bonds. When equities fall, bonds typically rally, providing ballast that stabilises the portfolio.

In 2022, that relationship broke sharply: aggressive central bank rate hikes drove both asset classes lower simultaneously, producing a joint drawdown that is particularly damaging to a strategy holding a large bond allocation.

Risk Parity returned −4.3% over the 2022 rate-hiking cycle, worse than Equal Weight (+1.5%) despite carrying significantly less equity risk. The regime that risk parity was, in principle, best positioned to weather proved to be one of its worst episodes in the sample.

The missing ingredient: Leverage

Institutional implementations of risk parity, most notably Bridgewater's All Weather strategy, apply leverage to scale the low-volatility portfolio up to a target return level, compensating for the return drag from bond overweighting.

In an unlevered setting, the lower return profile is not offset, and the strategy's risk-adjusted advantage over equities is structurally compressed. Our implementation is unlevered throughout, which is the appropriate choice for a like-for-like comparison with the other strategies, but it means Risk Parity is evaluated in a setting that partially disadvantages its intended design.

A leveraged version would likely produce a materially different risk-return profile, though at the cost of introducing leverage risk and financing costs beyond the scope of this study.

Qualification

These results should not be read as a dismissal of risk parity as a framework. Risk Parity provided the best downside protection during the COVID-19 crash (−13.7% versus Equal Weight's −25.8%) and consistently maintained lower peak-to-trough drawdowns across the full sample. For investors with lower loss tolerance, a shorter investment horizon, or a liability-matching objective, the drawdown characteristics may justify the return sacrifice. The puzzle, more precisely stated, is not that Risk Parity is a poor strategy in general, but that the specific conditions of 2015–2025 — a prolonged equity bull market followed by a simultaneous bond and equity drawdown in 2022 — were precisely those least suited to an unlevered risk parity implementation on a UK multi-asset universe.

5.3 Practical Implications

The results of this study carry several practical lessons for portfolio construction beyond the model-specific findings discussed above.

Model selection depends on investor objective

No single strategy dominates across all dimensions of the performance space. A practitioner's choice should be conditioned on their investment objective:

i. *Total return / risk-adjusted performance*: Equal Weight. No strategy in this study improved on it over the full sample.
ii. *Drawdown control / capital preservation*: Risk Parity. Its maximum

Key Points

- Black-Litterman returns and Sharpe ratio is entirely flat across $\tau \in \{0.01, 0.25\}$, confirming momentum views dominate the prior.
- Performance is inversely ordered by complexity: more inputs consistently produced worse results.
- Simplicity should be the rational default; complexity is justified only when input quality is high.

drawdown of −23.7% and rapid recovery from most dips suit investors with lower loss tolerance or liability-matching objectives, provided they accept the structural return drag from bond overweighting.

iii. *Systematic active tilts*: Black-Litterman, but only with carefully constructed views and well-calibrated parameters. A mechanically implemented momentum signal, as used here, does not reliably add value relative to the naive benchmark on this universe.

Turnover as a hidden performance drag

Transaction costs are absent from the gross performance figures reported throughout this study, but they are not absent from reality. MVO's average monthly turnover of 34.7% means that at every rebalancing, more than a third of the portfolio is being traded. Even modest transaction costs would reduce MVO's already weakest net returns further still, widening the gap with Equal Weight.

For a retail investor facing bid-ask spreads and brokerage fees that are multiples larger, MVO is effectively unimplementable without explicit cost controls such as turnover constraints or a minimum trade threshold.

By contrast, Equal Weight's 2.5% average monthly turnover imposes a negligible implementation cost, reinforcing its practical appeal beyond its favourable gross performance.

BL parameter sensitivity: prior domination confirmed

To assess the robustness of the Black-Litterman results to calibration choices, we ran a grid search over $\tau \in \{0.01, 0.05, 0.10, 0.25\}$ and $\delta \in \{1, 2, 3, 5\}$, yielding 16 independent backtests. The results are shown in Figure 12. The finding is striking: Returns and Sharpe ratio are **completely insensitive to** τ across the full range tested, for any fixed δ, returns vary less than 0.01 across all four values of τ. This is direct empirical evidence of prior domination: the momentum views Q are sufficiently large relative to the equilibrium prior $\mathbf{\Pi}$ that varying τ, which controls how much weight the prior receives, has no measurable effect.

The posterior is determined almost entirely by the views, regardless of how uncertain the prior is declared to be.

The risk-aversion coefficient δ has more impact: $\delta = 1$ produces a materially worse Sharpe (0.07) compared to $\delta \in \{2, 3, 5\}$, which all yield approximately 0.26. The degradation at $\delta = 1$ reflects how a lower risk-aversion coefficient produces smaller equilibrium returns ($\mathbf{\Pi} = \delta \mathbf{\Sigma} \mathbf{w}_m$), altering the posterior magnitude even when views dominate its direction. For $\delta \geq 2$, the model is robust to further increases in risk aversion.

The practical implication is direct: in this implementation, BL behaves like MVO with momentum expected returns regardless of τ. A practitioner wishing to genuinely exploit the Bayesian blending mechanism would need views smaller in magnitude relative to the equilibrium prior, or a substantially larger τ to reduce the prior's pull. Tuning τ within the conventional range offers no benefit here.

More information, worse performance

Perhaps the sharpest practical lesson of this study is that performance is inversely ordered by informational complexity. Equal Weight consumes no inputs beyond the asset universe itself. Risk Parity requires only the covariance matrix. MVO adds a vector of expected returns. Black-Litterman adds an equilibrium prior and investor views.

Yet the performance ranking runs in the opposite direction, with Equal Weight in first place. Each additional layer of input estimation introduces additional estimation error, and in a finite-sample setting

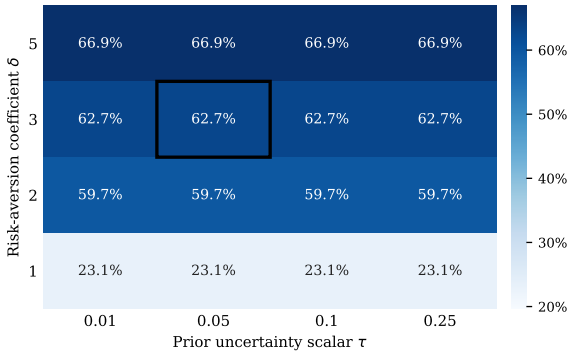

Figure 12. Total return (%) heatmap.
Black box marks the baseline configuration ($\tau = 0.05$, $\delta = 3$)

that error compounds faster than any informational benefit accrues.

This does not mean complex models are without value, but it means their value is conditional on the quality of their inputs. Practitioners should treat simplicity as the rational default and reserve complexity for settings where reliable return forecasts are genuinely available.

5.4 Limitations

Several limitations of this study should be borne in mind when interpreting its findings.

Single universe and time period

All results are drawn from a single 18-asset UK multi-asset portfolio over 2015-2025. This period is characterised by a prolonged equity bull market and a single rate-hiking episode: conditions that systematically favour Equal Weight's equity-heavy allocation and structurally disadvantage an unlevered Risk Parity strategy.

A different sample period or asset universe could produce materially different rankings. The finding that 1/N outperforms optimised strategies is consistent with the broader literature (DeMiguel et al.; 2009), but the magnitude of the gap is specific to this setting and should not be extrapolated.

No transaction costs

All performance metrics are reported gross of transaction costs. This overstates net performance for all strategies, but the distortion is highly uneven: MVO's average monthly turnover of 34.7% means that transaction costs would erode its returns far more severely than Equal Weight's 2.5%.

A realistic comparison would model costs explicitly. For example, as a fixed basis-point charge applied to each trade at every rebalancing date. The relative ordering of strategies could change once costs are accounted for, reinforcing the case against MVO even further.

Fixed estimation window

The 252-day rolling window used to estimate covariances and expected returns is a fixed design choice that is not optimised or validated. Shorter windows are more responsive to recent market conditions but introduce greater estimation noise; longer windows are more stable but slow to adapt to regime changes.

Key Points

- Results are specific to 18 UK assets over 2015–2025 and may not generalise to other settings.
- Performance is reported gross of transaction costs, which disproportionately disadvantages MVO.
- The asset universe is subject to survivorship bias due to ex-post selection of FTSE 100 constituents.

The sensitivity of results to window length is not explored here and represents a natural robustness check for future work.

Mechanical Black-Litterman views

The momentum signal used to construct BL views is a systematic, rules-based proxy rather than genuine investor views. The Black-Litterman framework was designed to incorporate analyst forecasts, macroeconomic judgments, and qualitative insights that a practitioner believes contain information beyond what the market has already priced in.

Using a mechanical momentum rule means BL is never operating as originally intended: it is effectively MVO with a pre-specified return signal, and its underperformance should be interpreted accordingly.

Results could differ substantially with views constructed from fundamentals, macro factors or genuine active research.

Survivorship bias

The 18 assets were selected ex-post as known, continuously-traded constituents of the FTSE 100 over the sample period.

Companies that were delisted, went bankrupt, or fell out of the index during 2015-2025 are excluded by construction, which upward-biases the equity returns in the universe and understates the true risk of holding UK large-cap equities.

A more rigorous analysis would construct the universe using point-in-time index membership data.

6 Conclusion

This study implemented and compared four portfolio allocation strategies across an 18-asset UK multi-asset universe over 2015-2025: an equal-weight benchmark, Markowitz mean-variance optimisation, Black-Litterman and Risk Parity. The central question was whether model sophistication translates into superior out-of-sample performance.

On the three research questions

The first question, whether optimisation models outperform 1/N out-of-sample, is answered clearly in the negative. Equal Weight delivered the highest total return (93.9%), the highest Sharpe ratio (0.36), and the highest Sortino ratio (0.47) of all four strategies, outperforming Black-Litterman by over 31 percentage points in cumulative return.

In answer to the second question, no optimised strategy improved on Equal Weight on a risk-adjusted basis; the ranking by Sharpe is EW (0.36), BL (0.26), RP (0.19), MVO (0.15). Model complexity consistently hurt rather than helped.

In answer to the third question, regime analysis reveals that no strategy provides universal protection: BL excelled during the Brexit shock (+20.3%), MVO was uniquely resilient during the COVID-19 crash (+3.6%) due to its concentration in safe havens and Risk Parity, despite its theoretical all-weather appeal, was among the worst performers in the 2022 rate-hiking episode (−4.3%) when the equity-bond correlation that underpins it broke down.

The central finding

The most striking pattern is that performance is inversely ordered by informational complexity: the strategy consuming the fewest inputs performed best, and each additional layer of estimation made things worse.

This is not a failure of portfolio theory. It is a consequence of estimation error in finite samples: optimisers amplify input noise into unstable, concentrated portfolios that underperform simpler alternatives out-of-sample.

Black-Litterman partially addresses this through its equilibrium prior, but in this implementation the prior is dominated by mechanical momentum views, reducing BL to MVO with a different signal.

Risk Parity avoids expected return estimation entirely and provides meaningful downside protection, but its structural bond overweight drags performance in equity-dominated environments.

Future work

Several natural extensions follow from the limitations of this study.

Modelling transaction costs explicitly would provide a more realistic comparison, particularly for MVO.

Constructing the asset universe from point-in-time index membership data would eliminate survivorship bias. Replacing mechanical momentum views with fundamentals-based or macro-driven signals would allow BL to operate as intended.

Applying leverage to Risk Parity would bring it closer to its institutional implementation.

Finally, extending the comparison to include Hierarchical Risk Parity and constrained MVO would complete the picture of how regularisation and structural constraints affect the estimation error problem.

7 Declarations

Author Contributions

R.d.S. led the project, coordinated all research activities, implemented the Risk Parity model and contributed to Sections 2 and 4.

J.H. built the backtesting engine, implemented the Black-Litterman model and wrote Section 3.

S.K. implemented the Equal Weight benchmark and the Mean-Variance Optimisation models and wrote Section 2.

F.C. implemented the MVO implementation with S.K. and wrote Section 5.

A.I. implemented Risk Parity with R.d.S and wrote Section 1.

All authors reviewed and approved the final manuscript.

Code Availability

All code, backtesting infrastructure and figure generation scripts are publicly available at (Surrey Capital Research; 2026).

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